

FortiGate on Containerlab (via vrnetlab)

FOR
DUMMIES

FortiGate is not a container. It ships as a qcow2 disk image — a full VM — so Containerlab runs it through [vrnetlab](#), which wraps a vendor VM in a container that QEMU boots at start-up.

That has two consequences worth understanding before you start:

1. **There is no image to pull.** `vrnetlab/vr-fortios:7.0.9` does not exist on any registry. You build it locally from the qcow2. `02-deploy-lab.sh` checks for it and stops with this message rather than letting Containerlab try a registry and fail with a confusing “pull access denied”.
2. **It needs hardware virtualization.** QEMU inside the container wants `/dev/kvm`. On a VM (which this lab host is), that means *nested* virtualization has to be switched on at the hypervisor. Without it FortiGate either refuses to boot or falls back to pure emulation and takes many minutes per node.

The short version

```
./lab fortigate           # or: ./scripts/03-build-vrnetlab.sh
```

That one command does everything below: checks `/dev/kvm` and tells you exactly which hypervisor setting to change if it's missing, installs build dependencies, clones or updates `vrnetlab`, finds your `.qcow2`, builds, reports the image tag it actually produced, and offers to update your topology files to match that tag. It's safe to re-run.

`setup.sh` also offers it, but only when one of your topologies actually references a `vrnetlab/` image — the cEOS-only path never sees the prompt.

The rest of this document is what that script does, for when you want to do it by hand or something goes sideways.

1. Check nested virtualization first

Do this before downloading anything — if it's off, nothing else matters.

```
ls -l /dev/kvm           # must exist
grep -Ec '(vmx|svm)' /proc/cpuinfo  # must be > 0
```

If `/dev/kvm` is missing:

- vSphere / ESXi — power the VM off, then Edit Settings → expand CPU → tick Hardware virtualization: Expose hardware assisted virtualization to the guest OS. Power back on. (docs/vcenter-vm-setup.md builds the VM without this, since cEOS doesn't need it.)
- Proxmox — set the VM's CPU type to `host` and make sure nesting is enabled on the Proxmox host. See docs/proxmox-vm-setup.md.

2. Get the qcow2

Team Drive (internal mirror):

https://drive.google.com/drive/folders/1kBDv_xgv4T4NQJfWZtLKkCnYbmcfs9KU?usp=drive_link → ContainerLab/Fortigate Images/fortinet-FGT-v7.0.9-build0444.qcow2

Or from the vendor, if you'd rather pull it yourself or need a different build:

<https://support.fortinet.com> → Download → VM Images → select the KVM image for your version.

Either way you need a valid Fortinet entitlement to use the image — the mirror is a convenience for people who already have one, not a way around licensing.

Copy it to the lab host:

```
scp fortinet-FGT-v7.0.9-build0444.qcow2 <you>@<VM-IP>:~/
```

3. Build the container image

```
sudo apt-get install -y make git qemu-kvm
git clone https://github.com/hellt/vrnetlab.git ~/vrnetlab
cd ~/vrnetlab
ls                                     # find the FortiGate directory for your vrnetlab version
```

The directory name has changed across vrnetlab releases — current trees use `fortinet_fortigate/`, older ones `fortios/`. Use whichever your clone has:

```
cd ~/vrnetlab/fortinet_fortigate      # or fortios/ on older trees
cp ~/fortinet-FGT-v7.0.9-build0444.qcow2 .
make
```

The build boots the VM once to prepare it, so it takes a few minutes.

4. Match the tag to your topology

This is the step people miss. vrnetlab's image tag has also changed across versions — newer trees produce `vrnetlab/fortinet_fortigate:<version>`, older ones `vrnetlab/vr-fortios:<version>`. Check what you actually got:

```
docker images | grep -i forti
```

Then make `topologies/memory-test.clab.yml` agree with it:

```
kinds:
  fortinet_fortigate:
    image: vrnetlab/vr-fortios:7.0.9      # <- must match `docker images`
```

A mismatch here produces exactly the “not available locally” stop from `02-deploy-lab.sh`, which is the intended behaviour — it's telling you the truth.

5. Deploy

```
./lab topo memory-test.clab.yml  
./lab deploy
```

FortiGate nodes boot far more slowly than cEOS — allow several minutes before the management interface answers. `./lab status` shows when the containers are up; the VM inside takes longer still.

Default credentials on a fresh FortiGate VM are `admin` with an **empty** password, and it forces a password change on first login. That's a different account from the cEOS nodes' `admin/admin`.

Resource cost

A FortiGate VM node is much heavier than a cEOS node — roughly 1–2 GB RAM each plus QEMU overhead, against ~0.5–1 GB for cEOS. `memory-test.clab.yml` runs two of them alongside four cEOS nodes and four small containers; budget around 8–10 GB of the VM's 32 GB, and destroy the lab when you stop for the day.